

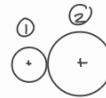
Système contrain

Courrie / engrenage

$$\Theta_1 R_1 = \Theta_2 R_2$$

$$\omega_1 R_1 = \omega_2 R_2$$

$$\alpha_1 R_1 = \alpha_2 R_2$$



$$V_1 = V_2$$

$$L_1 = L_2$$

$$a_{t1} = a_{t2}$$

Mouvement circulaire

$$\vec{a}_{t\text{total}} = \vec{a}_t + \vec{a}_c$$

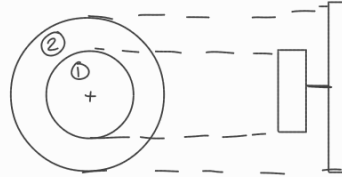
$$a_t = \frac{\Delta v}{\Delta t} \quad a_c = \frac{v^2}{R}$$

Roues concentrique

$$\Theta_1 = \Theta_2$$

$$\omega_1 = \omega_2$$

$$\alpha_1 = \alpha_2$$



Exercice #2

①	$\alpha = ?$	②
$\Theta_1 = 0$		$\Theta_2 = ?$
$\omega_1 = 78 \text{ RPM}$		$\omega_2 = 0$
$t_1 = 0$		$t_2 = 30 \text{ s}$

$$\Theta_2 = 0 + \left(78 \cdot \frac{2\pi}{60}\right) (30 - 0) + \frac{1}{2} \alpha (30 - 0)^2$$

$$0 = \left(78 \cdot \frac{2\pi}{60}\right) + \alpha (30 - 0)$$

$$\alpha = -0.27 \text{ rad/s}^2$$

$$\Theta_2 = 122.522 \text{ rad}$$

$$\Theta_2 = \frac{122.522}{2\pi} = 19.5 \text{ tour}$$

Exercice 4

MCUA

$$\Theta_2 = \Theta_1 + \omega_1 (t_2 - t_1) + \frac{1}{2} \alpha (t_2 - t_1)^2$$

$$\omega_2 = \omega_1 + \alpha (t_2 - t_1)$$

①	$\alpha = ?$	②	③
$\Theta_1 = 0$		$\Theta_2 = ?$	$\Theta_3 = \Theta_2 + 60$
$\omega_1 = 0$		$\omega_2 = 10 \text{ RPS}$	$\omega_3 = 15 \text{ rps}$
$t_1 = 0$		$t_2 = ?$	$t_3 = t_2 + t$

2 → 3

$$\Theta_2 + 60 \cdot 2\pi = \Theta_2 + (10 \cdot 2\pi) (t_2 + t - t_2) + \frac{1}{2} \alpha (t_2 + t - t_2)^2$$

$$a) 15 \cdot 2\pi = 10 \cdot 2\pi + \alpha (t_2 + t - t_2)$$

$$\alpha = 6.54498 \text{ rad/s}^2$$

$$t = 4.8 \text{ s}$$

①	$\alpha = 6.54498 \text{ rad/s}^2$	②
$\Theta_1 = 0$		$\Theta_2 = 60 \cdot 2\pi$
$\omega_1 = 0$		$\omega_2 = ?$
$t_1 = 0$		$t_2 = ?$

$$60 \cdot 2\pi = 0 + 0 \cdot (t_2 - 0) + \frac{1}{2} \alpha (t_2 - 0)^2$$

$$\omega_2 = 0 + \alpha (t_2 - 0)$$

$$\omega_2 = 70.2481 \text{ rad/s}$$

$$t_2 = 10.7 \text{ sec}$$

①	$\alpha = 6.54498$	②
$\Theta_1 = 0$		$\Theta_2 = ?$
$\omega_1 = 0$		$\omega_2 = 10 \cdot 2\pi$
$t_1 = 0$		$t_2 = ?$

$$\Theta_2 = 0 + 0 (t_2 - t_1) + \frac{1}{2} \alpha (t_2 - 0)^2$$

$$10 \cdot 2\pi = 0 + \alpha (t_2 - 0)$$

$$\Theta_2 = 301.593 \text{ rad} / 2\pi \rightarrow 48 \text{ tour}$$

$$t_2 = 9.6 \text{ sec}$$

Exercise #7

$$R = 300\text{m}$$

$$V_A = 15\text{m/s}$$

$$V_B = 20\text{m/s}$$

$$a_t = cte = \alpha \cdot R$$

$\downarrow$
cte

$$V_A = \omega_A \cdot R \rightarrow \omega_A = 0,05$$

$$V_B = \omega_B \cdot R \rightarrow \omega_B = 0,066667$$

$$\theta_2 = \theta_1 + \omega_1 (t_2 - t_1) + \frac{1}{2} \alpha (t_2 - t_1)^2$$

$$\omega_2 = \omega_1 + \alpha (t_2 - t_1)$$

①	$\alpha = ?$	②
$\theta_1 = 0$		$\theta_2 = \frac{\pi}{2}$
$\omega_1 = 0,05$		$\omega_2 = 0,066667$
$t_1 = 0$		$t_2 = ?$

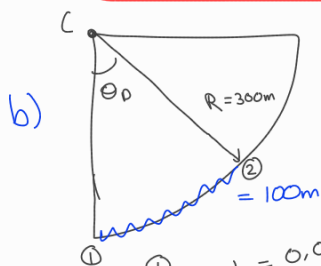
$$\frac{\pi}{2} = 0 + 0,05 (t_2 - 0) + \frac{1}{2} \alpha (t_2 - 0)^2$$

$$0,066667 = 0,05 + \alpha (t_2 - 0)$$

$$t_2 = 26,9279\text{ sec}$$

$$\alpha = 0,000619\text{ rad/sec}^2$$

$$a_t = 0,000619 \cdot 300 = 0,186\text{ m/s}^2$$



$$\theta_D = \frac{L}{R} = \frac{100}{300}$$

$$\alpha = 0,000619$$

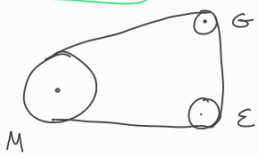
①	②
$\theta_1 = 0$	$\theta_2 = \frac{100}{300}$
$\omega_1 = 0,05$	$\omega_2 = ?$
$t_1 = 0$	$t_2 = ?$

$$t_2 = 6,412\text{ sec}$$

$$\omega_2 = 0,054\text{ rad/sec} \Rightarrow V_2 = \omega_2 \cdot R = 0,054 \cdot 300 = 16,19\text{ m/s}$$

$$a_c = \frac{V^2}{R} = \frac{(16,19)^2}{300} = 0,87372\text{ m/s}^2$$

Exercice #11



$$D_m = 20 \text{ cm} = 0,2 \text{ m}$$

$$D_g = 8 \text{ cm} = 0,08 \text{ m}$$

a) $\omega_m R_m = \omega_E R_E$

$$\omega_E = 2 \omega_m$$

$$\omega_m R_m = 2 \omega_m R_E$$

$$R_E = \frac{R_m}{2} = 0,05 \text{ m} = 5 \text{ cm}$$

b) M.C.U.A. $\alpha = \text{cte}$

①	$\alpha = ?$	②
$\theta_1 = 0$		$\theta_2 = ?$
$\omega_1 = 3000 \text{ RPM}$		$\omega_2 = 40000 \text{ RPM}$
$t_1 = 0$		$t_2 = 4 \text{ s}$

b-2)

$$\theta_{2m} = 0 + \left(3000 \cdot \frac{2\pi}{60}\right)(4-0) + \frac{1}{2} \alpha (4-0)^2$$

$$\left(40000 \cdot \frac{2\pi}{60}\right) = \left(3000 \cdot \frac{2\pi}{60}\right) + \alpha (4-0)$$

$$\alpha = 26,18 \text{ rad/s}^2$$

$$\theta_2 = 1466,08 \text{ rad}$$

$$a_t = \alpha_m \cdot R_m = 26,18 \cdot 0,1 = 2,618 \text{ m/s}^2$$

↓
acceleration
courroie